Deciding Whether a Relation Is a Function
Answer Key

Grade 8 / Pre-Algebra   Function Concepts and Comparisons

Quick Answers

1. 5, —	4. 3, 3	7. 4, 7, —	10. 214, —
2. —	5. 12, —	8. 4	
3. —, 16	6. 4, —	9. —, 5	

Curriculum

Level: Grade 8 / Pre-Algebra   Function Concepts and Comparisons
8.F.A.1–8.F.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 22 tagged checks.

1. Mapping diagram: inputs 2, 4, 6.

Step 1 (part a). Follow the single arrow leaving the input 4. It lands on $\boxed{5}$
Step 2 (part b). Count the arrows: three. Count the different inputs: 2, 4 and 6, also three. So $\boxed{3 = 3}$
Step 3. The counts are equal, so no input was used twice: this *is* a function.
Note: two inputs share the output 5, and that is allowed. The rule restricts inputs, not outputs.

2. Mapping diagram: inputs 1, 3, 7.

Step 1. Count the arrows: four. Count the different inputs: 1, 3 and 7, only three.
Step 2. $\boxed{3 < 4}$ — fewer inputs than arrows means one input must be carrying two arrows.
Step 3. The input that spoils it is 3: it sends to 6 and to 8. Because that one input has two outputs, the relation is *not* a function.
Note: finding the offending input is the proof. "Not a function" without naming it is only half an answer.

3. The set $\{(1, 4), (2, 7), (3, 10), (4, 13)\}$.

Step 1 (part a). Four points are plotted, and their first coordinates are 1, 2, 3, 4 — four different ones. So $\boxed{4 = 4}$, and the relation is a function.
Step 2 (part b). Each output is three more than the last while each input goes up by one, so the rule is $y = 3x + 1$. Check it on a known pair: $3(2) + 1 = 7$ ✓
Step 3. At the input 5: $3(5) + 1 = \boxed{16}$
Note: the rule is what lets the relation be written $f(x) = 3x + 1$. A relation that is not a function has no such single rule.

4. The graph of $y = |x|$.

Step 1 (part a). At the input -3 the left branch is at height $\boxed{3}$, and at the input 3 the right branch is at the same height, 3.
Step 2 (part b). Every vertical line meets this V exactly once, so it passes the vertical line test and is a function.
Step 3 (part c) — manual review. *Full credit says:* the definition restricts each *input* to one output. Here the inputs -3 and 3 are different inputs that happen to share an output, so no input has been given a second value and nothing is broken. If instead one input — say 3 — had outputs 3 and -3, then asking "what is the output at 3?" would have two answers, and the vertical line $x = 3$ would cut the graph twice.
Note: outputs may repeat; inputs may not. Almost every wrong answer on this topic comes from swapping those two.

5. Hourly temperature table.

Step 1 (part a). Reading down the hour row to 3 and across gives $\boxed{12}$, and the plotted point at hour 3 sits at the same height.
Step 2 (part b). Five readings, five different hours: $\boxed{5 = 5}$
Step 3. The counts match, so "hour $\to$ temperature" is a function — even though the readings at hours 4 and 5 are both 15. Repeated outputs again, and again harmless.
Note: a time-and-measurement table is a function for a physical reason: the thermometer can only have one reading at a given moment.

6. The circle of radius 5.

Step 1 (part a). On the circle, $x^2 + y^2 = 25$, so at the input 3 the outputs satisfy $y^2 = 25 - 9 = 16$. The upper one is $\boxed{4}$ and the lower one is -4.
Step 2 (part b). A function permits one output per input; this graph supplies two. So $\boxed{1 < 2}$
Step 3 (part c) — manual review. *Full credit says:* the circle fails the vertical line test, so it is not a function. A witness is the vertical line $x = 3$, which crosses the circle at $(3, 4)$ and at $(3, -4)$ — one input with two outputs. Any vertical line strictly between $x = -5$ and $x = 5$ would do; the two edge lines $x = \pm 5$ touch only once and are *not* witnesses.
Note: name a specific failing line. "It looks like it fails" is not an argument, and the two tangent lines show that not every vertical line works.

7. Books read by four students.

Step 1 (part a). Ben's bar is at $\boxed{4}$
Step 2 (part b). The tallest bars are Ana's and Cy's, both at $\boxed{7}$
Step 3 (part c). Reversed, the inputs are the book totals. There are still four arrows, but only three different totals — 7, 4 and 2 — so $\boxed{3 < 4}$
Step 4. The input 7 now sends to both Ana and Cy, so the reversed relation is *not* a function.
Note: this is the whole point of the problem. "Student $\to$ books" is a function and "books $\to$ student" is not, so being a function is a property of the *direction*, not of the data.

8. Solving for k so the diagram is a function.

Step 1. The input 2 has two arrows. That is only allowed if both arrows land on the same output — a function may not give one input two different values, but two arrows to one value is just the same value drawn twice.
Step 2. Set the two outputs equal: $3k = 12$.
Step 3. Divide both sides by 3: $\boxed{k = 4}$
Note: with $k = 4$ both arrows from 2 point at 12, which is the single output that input is allowed. Any other k makes the diagram fail.

9. Four plotted points.

Step 1 (part a). The points are $(-2, 1)$, $(0, 3)$, $(2, 5)$ and $(0, -1)$. Four points, but the first coordinates are only -2, 0 and 2 — three different inputs. So $\boxed{3 < 4}$, and the relation is not a function: the input 0 has outputs 3 and -1.
Step 2 (part b). Checking the point whose input is 2 against $y = x + 3$: $2 + 3 = \boxed{5}$ ✓, which matches the plotted point.
Step 3. Deleting $(0, -1)$ leaves $(-2, 1)$, $(0, 3)$ and $(2, 5)$, all on $y = x + 3$, with three points and three different inputs — a function.
Note: deleting $(0, 3)$ instead also repairs the count, but the survivors no longer lie on one line. Both answers satisfy the definition; only one keeps the pattern.

10. Students and lockers.

Step 1 (part a). Dev's arrow lands on locker $\boxed{214}$
Step 2 (part b). Four arrows, four different students: $\boxed{4 = 4}$, so ”student → locker” is a function.
Step 3 (part c) — manual review. *Full credit says:* reversing the arrows makes the lockers the inputs. All four locker numbers are different, so each one has exactly one arrow leaving it and the reversed relation is also a function. That happens precisely because no two students were given the same locker — no output was repeated in the original. The single change that would break it: give two students the same locker number (say, put Cy in 214 as well). The original would still be a function — each student would still have one locker — but reversed, the input 214 would send to two students.
Note: a relation whose reverse is also a function is called *one-to-one*. Students do not need the word, but this is the idea it names, and it is exactly what problem 7 was missing.